Szymon Sarnowicz

szymon@sarnowicz.net | 425-802-6406 | linkedin.com/in/szymonsarnowicz | github.com/zymoncone | szymon.wiki

Experience

Readiscover.app – Founding Technical Lead | *Jan 2026 – present*

- Built full-stack semantic search platform for long-form text retrieval. Engineered adaptive chunking strategy and deployed multi-layer caching: GCS for persistent storage, Redis for sub-second cache access with optimized TTL.
- Implemented LLM query reformulation (Gemini 2.5 Flash at temperature 0.0 for deterministic output) before embedding. System design prioritizes trustworthiness: returns only continuous verbatim passages, eliminating hallucination risk (vs. 43.6% for GPT-5.1).
- Conducted rigorous ablation testing across chunking parameters; evaluated against GPT-5.1 on 50 test questions achieving 34% top-3 accuracy with zero hallucinations and superior proximity to correct passages. Results published at University of Michigan and now deployed as production platform.

General Motors – Software Engineer II | *Feb 2023 – present*

- Owned end-to-end redesign of safety-critical fault detection system in C/C++, implementing sensor data processing, signal filtering algorithms, and state machine logic for fault progression. Architected persistent state management to prevent critical system failures, helping avoid ~$480M in potential recalls.
- Architected and deployed production-level diagnostic system from concept to deployment; designed state monitoring logic, implemented condition detection algorithms, and integrated alert routing protocols across distributed services. Increased system reliability by refactoring fault-response behavior across microservices and developing a Python testing framework for safety-critical validation.
- Mentored 2-3 junior engineers on safety-critical systems design and code best practices, helping them ship production code for fault detection and diagnostics pipelines.

Data4Good – University of Michigan – Lead Developer, Hangul Project | *Dec 2023 – Dec 2024 (Volunteer)*

- Developed Svelte web application for United Nations document summarization using Apache Tika for HTML parsing and BART for NLP extraction. Presented product to UN representatives for feedback. Redesigned API pipeline into multiple stages with exponential backoff, increasing success rate from 53% to 92%.

General Motors – Program Manager, Autonomous Systems (Ultra Cruise) | *Jul 2021 – Feb 2023*

- Tripled data collected per sprint by automating data-request workflows, improving visibility for perception and autonomy engineering teams. Enabled faster iteration cycles for sensor fusion and real-time decision-making pipelines in autonomous vehicle development.
- Authored comprehensive troubleshooting guides and diagnostics workflows that enabled cross-functional teams to independently resolve camera, radar, and LiDAR system issues. Bridged hardware integration and software systems, improving sensor calibration diagnostics and reducing development cycle time.

General Motors – Software Engineer (Rotational Program) | *Sep 2019 – Jul 2021*

- Automated calibration data management processes across embedded systems, eliminating 1 hour of manual data entry work per day. Gained foundational experience in embedded systems, real-time data handling, sensor calibration, and automotive software development practices.

Education

Master's in Applied Data Science, University of Michigan, summa cum laude | Dec 2025
B.S. Mechanical Engineering, University of Washington, cum laude | Jun 2019

Technical Skills & Awards

Languages: Python, C, C++, JavaScript

Systems: C++ for embedded systems, real-time signal processing, state machines

ML/AI: LLMs, semantic search, embeddings, NLP, RAG

Cloud & Infrastructure: GCP (Vertex AI, Cloud Run), AWS (Lambda, DynamoDB), Docker, Redis

Tools: React, FastAPI, Git, Linux

Best Cybersecurity Hack, MHack16 (350+ participants) | Dec 2023